

Lab 3 - Discrete Monte Carlo

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PART 1: DISCRETE DISTRIBUTION FUNCTIONS SECTION

```
set.seed(67) # after all why not
nb_values <- 0:5
nb_pdf <- dnbinom(nb_values, size = 3, prob = 0.8)
data.frame(x = nb_values, PDF = nb_pdf) #0-5 s=3 p=.8
```

##	x	PDF
## 1	0	0.51200000
## 2	1	0.30720000
## 3	2	0.12288000
## 4	3	0.04096000
## 5	4	0.01228800
## 6	5	0.00344064

```
qgeom(0.30, prob = 0.02) #30% of a geometric distribution with success probability 0.02
```

```
## [1] 17
```

```
pbinom(8, size = 20, prob = 0.4) #P(X = 8) for Binomial(20, 0.4)
```

```
## [1] 0.5955987
```

```
pois_realizations <- rpois(50, lambda = 10) #50 realizations + 10 lambda
data.frame(Realization = pois_realizations) #50 Realizations
```

##	Realization
## 1	13
## 2	9
## 3	6
## 4	10
## 5	11
## 6	10
## 7	9
## 8	8
## 9	8
## 10	11
## 11	14
## 12	6
## 13	8
## 14	11
## 15	7
## 16	8
## 17	10

```
## 18      9
## 19     10
## 20     15
## 21     12
## 22     12
## 23     11
## 24     14
## 25     11
## 26     13
## 27     15
## 28      7
## 29      8
## 30     10
## 31      6
## 32      7
## 33      9
## 34      7
## 35     12
## 36     12
## 37      9
## 38     11
## 39     11
## 40      9
## 41     15
## 42      6
## 43     10
## 44     11
## 45     10
## 46     13
## 47      8
## 48      5
## 49     10
## 50     11
```

PART 2: MONTE CARLO SIMULATION - THEORETICAL COMPARISON SECTION

```
#1
binom_sim <- rbinom(100000, size = 100, prob = 0.3)
var(binom_sim)#Monte Carlo Binomial variance
```

```
## [1] 21.07549
```

```
binom_var_theoretical <- 100 * 0.3 * (1 - 0.3)
binom_var_theoretical#Theoretical Binomial variance
```

```
## [1] 21
```

```
var(binom_sim) - binom_var_theoretical #Difference
```

```
## [1] 0.07548822
```

```
#2
x_sim <- rpois(100000, lambda = 2)
y_sim <- rgeom(100000, prob = 0.5)
```

```

sum_sim <- x_sim + y_sim
sum_mean_mc <- mean(sum_sim)
sum_mean_theoretical <- 2 + (1 - 0.5) / 0.5 #sum mean mathematically
sum_mean_mc #Monte Carlo E[X+Y]

```

```
## [1] 2.99661
```

```
sum_mean_theoretical #Theoretical E[X+Y]
```

```
## [1] 3
```

```
sum_mean_mc - sum_mean_theoretical #Difference
```

```
## [1] -0.00339
```

```
#3
```

```
nb_sim <- rbinom(100000, size = 5, prob = 0.65) #rand nb s=5 p=.65
```

```
p_gt_6_mc <- mean(nb_sim > 6)
```

```
p_gt_6_theoretical <- 1 - sum(sapply(0:6, function(k) { choose(k + 5 - 1, k) * (1 - .65)^k * .65^5})) #
```

```
p_gt_6_mc #Monte Carlo
```

```
## [1] 0.04923
```

```
p_gt_6_theoretical #Theoretical
```

```
## [1] 0.05014271
```

```
p_gt_6_mc - p_gt_6_theoretical #Difference
```

```
## [1] -0.0009127104
```

PART 3: COMPLEX MONTE CARLO SIMULATION SECTION

```
#1
```

```
humidity <- rbinom(100000, size = 5, prob = 0.3) #Cond 1
```

```
rain_prob <- humidity / 6 #Cond 2
```

```
will_rain <- rbinom(100000, size = 1, prob = rain_prob) #Cond 2(Rain is True)
```

```
sensors_working <- matrix(rbinom(100000 * 3, size = 1, prob = 0.7), ncol = 3) #Cond 3
```

```
detection <- will_rain * (rowSums(sensors_working) > 0) #Cond 3
```

```
rain_detector_sim <- data.frame(detection = detection)
```

```
mean(rain_detector_sim$detection) #Probability rain detector detects rain
```

```
## [1] 0.24368
```

```
#2
```

```
day_types <- rep(1:7, 20) # 140 days
```

```
simulate_accidents <- function() {
```

```
  rain_days <- rbinom(140, 1, 0.25) #.25 rainy days in 140
```

```
  lambda_cars <- ifelse(day_types >= 6, ifelse(rain_days == 1, 50, 300), 200) #1-5 weekday, 67 Weekend :)
```

```
  cars <- rpois(140, lambda_cars) #CPD
```

```
  accident_prob <- ifelse(rain_days == 1, 0.01, 0.001) #prob acc
```

```
  accidents <- rbinom(140, size = cars, prob = accident_prob) #apd
```

```
  sum(accidents) #ovr 140
```

```
} #fct form to simulate
```

```
accidents_total <- replicate(100000, simulate_accidents())
```

```
accidents_sim <- data.frame(total_accidents = accidents_total)
```

```
accidents_mean <- mean(accidents_sim$total_accidents)
```

```
accidents_sd<-sd(accidents_sim$total_accidents)
accidents_mean#mean
```

```
## [1] 78.9949
```

```
accidents_sd#sd
```

```
## [1] 11.82837
```

```
hist(accidents_sim$total_accidents, freq=F, main = "Histogram of Total Accidents with Normal Curve", xlab="Total Accidents", ylab="Density", col="lightblue", border="black", las=1)
lines(dnorm(0:200, mean=mean(accidents_sim$total_accidents), sd=sd(accidents_sim$total_accidents)), col="red", lty=1)
```

Histogram of Total Accidents with Normal Curve

